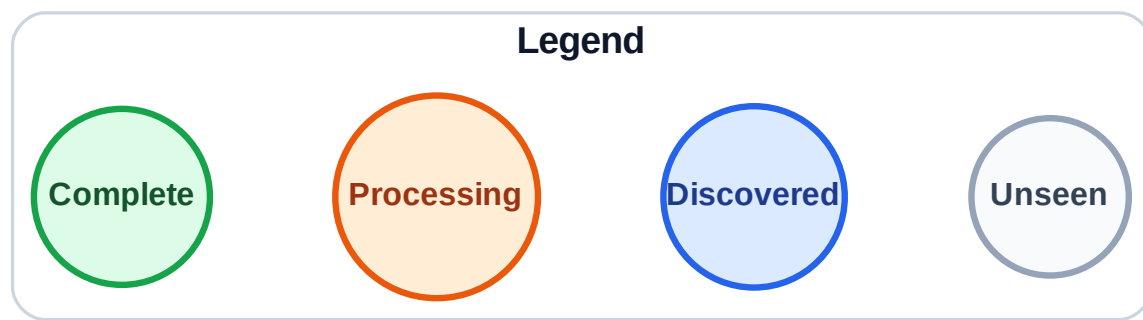
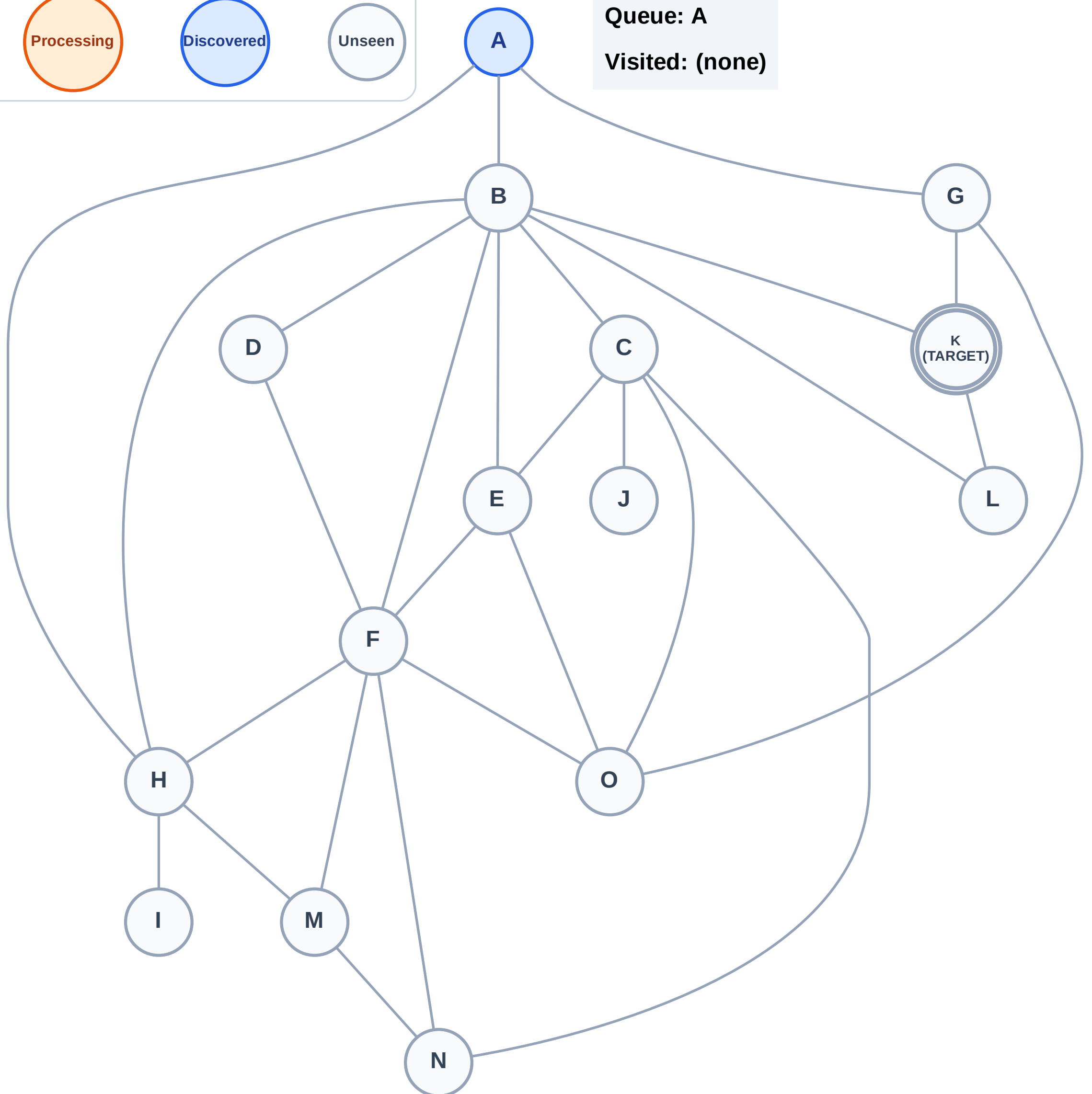


BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



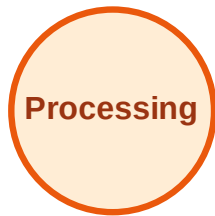
Queue: A
Visited: (none)



BFS Traversal

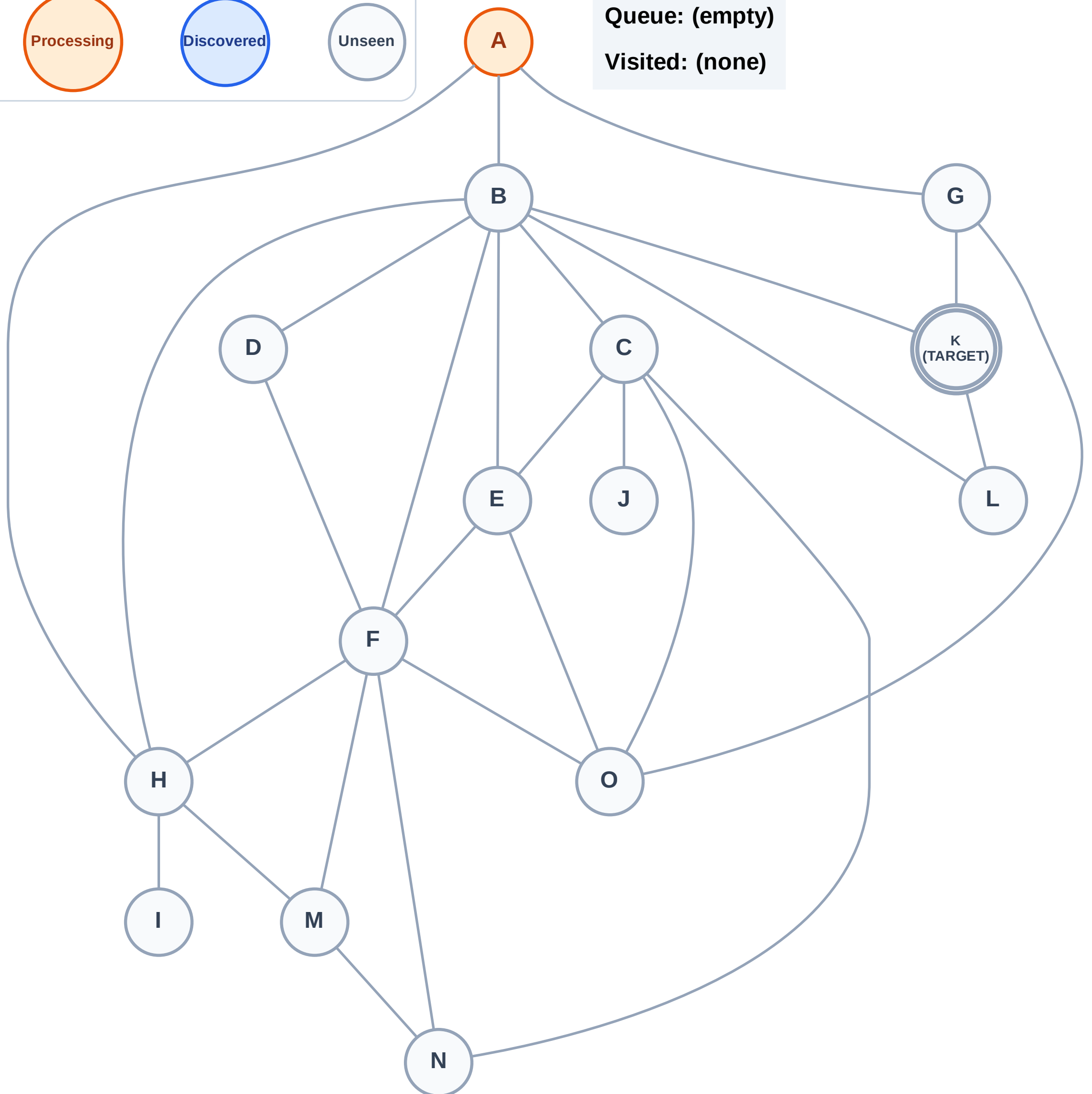
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



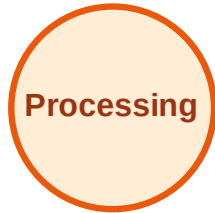
BFS Traversal

Step 3: Discover B, G, H from A

Legend



Complete



Processing



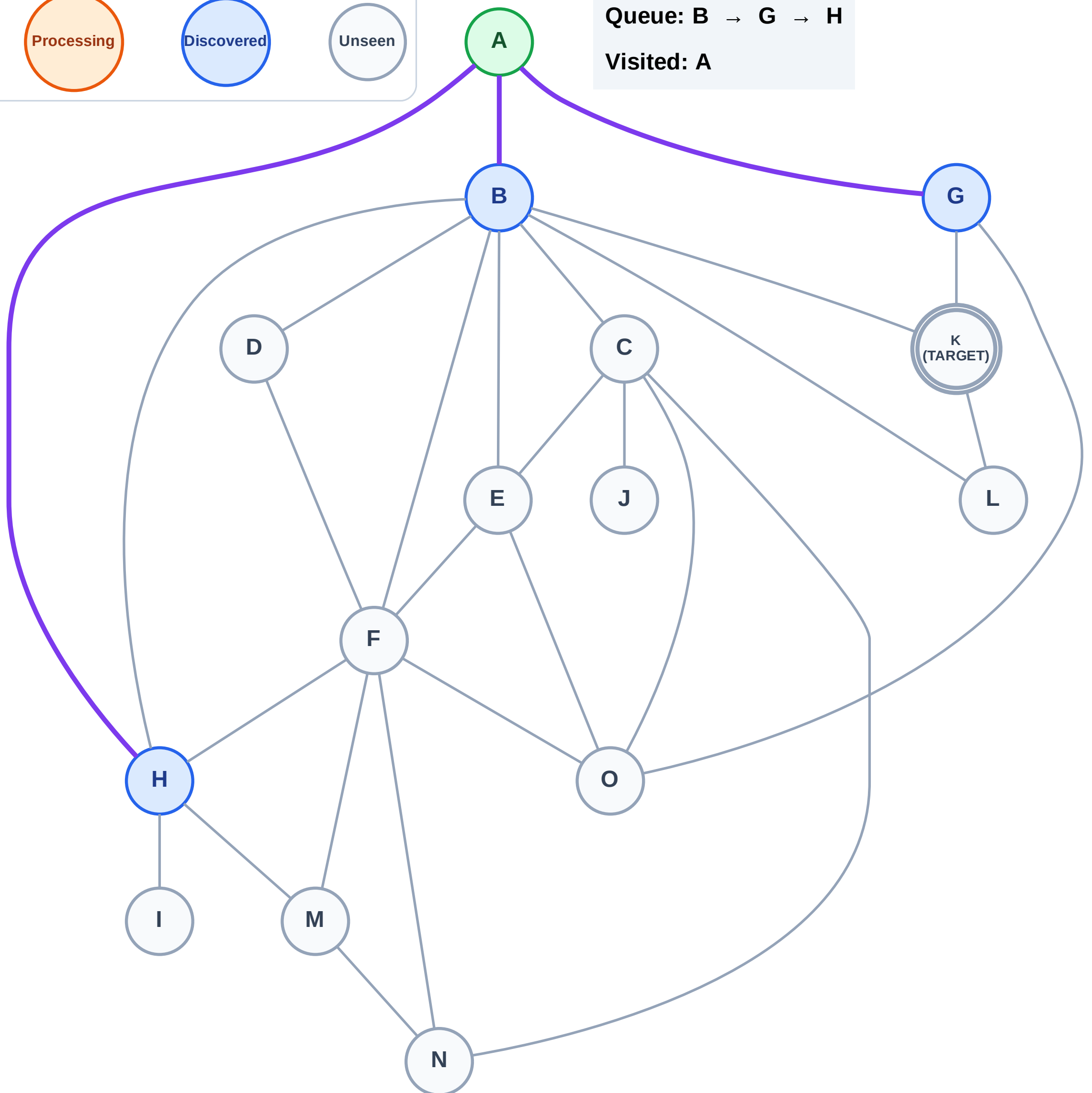
Discovered



Unseen

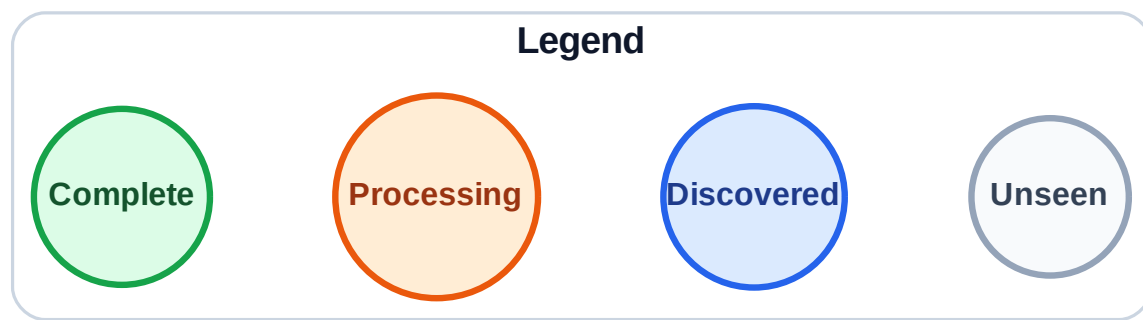
Queue: B → G → H

Visited: A



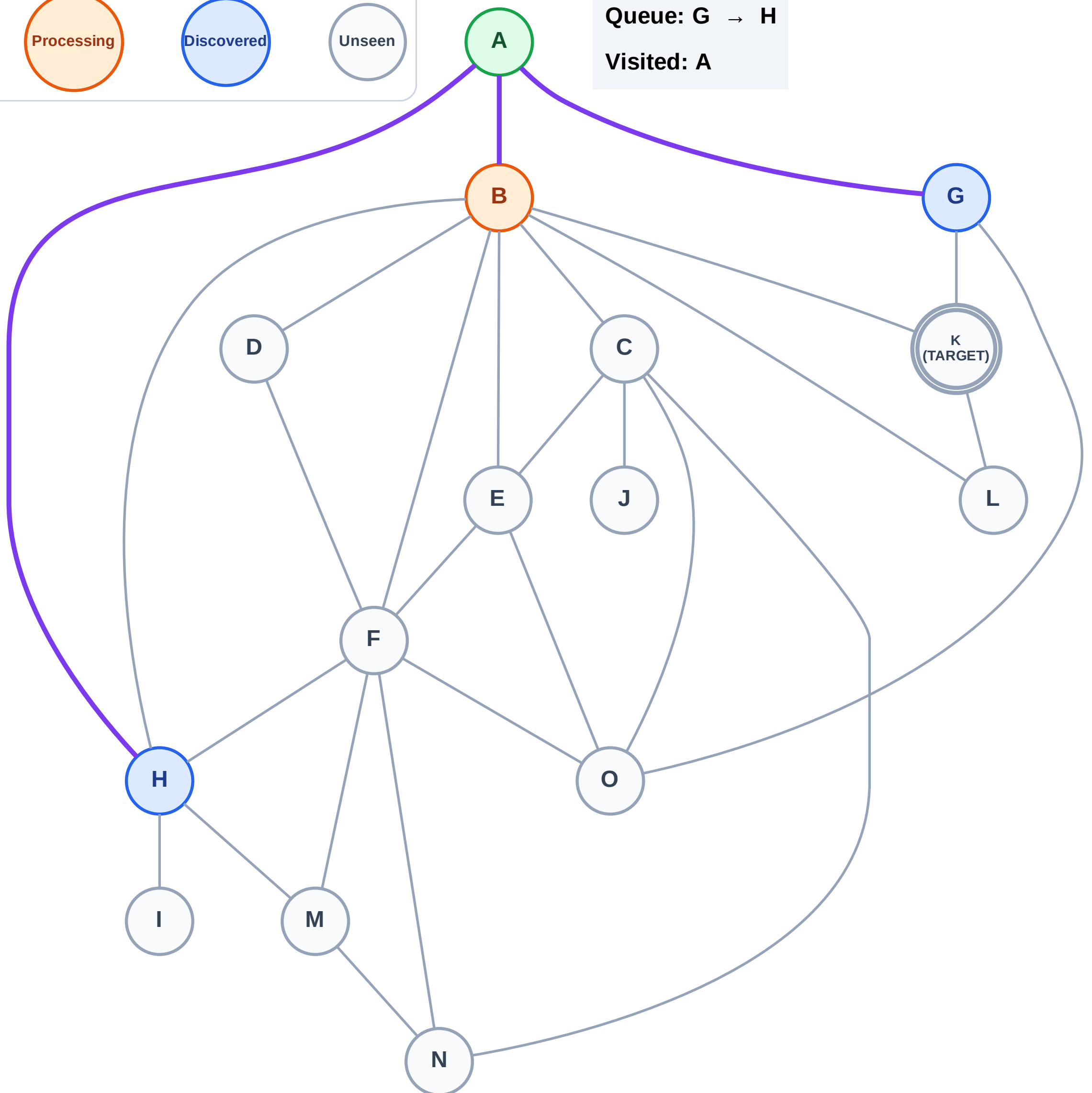
BFS Traversal

Step 4: Process node B



Queue: G → H

Visited: A



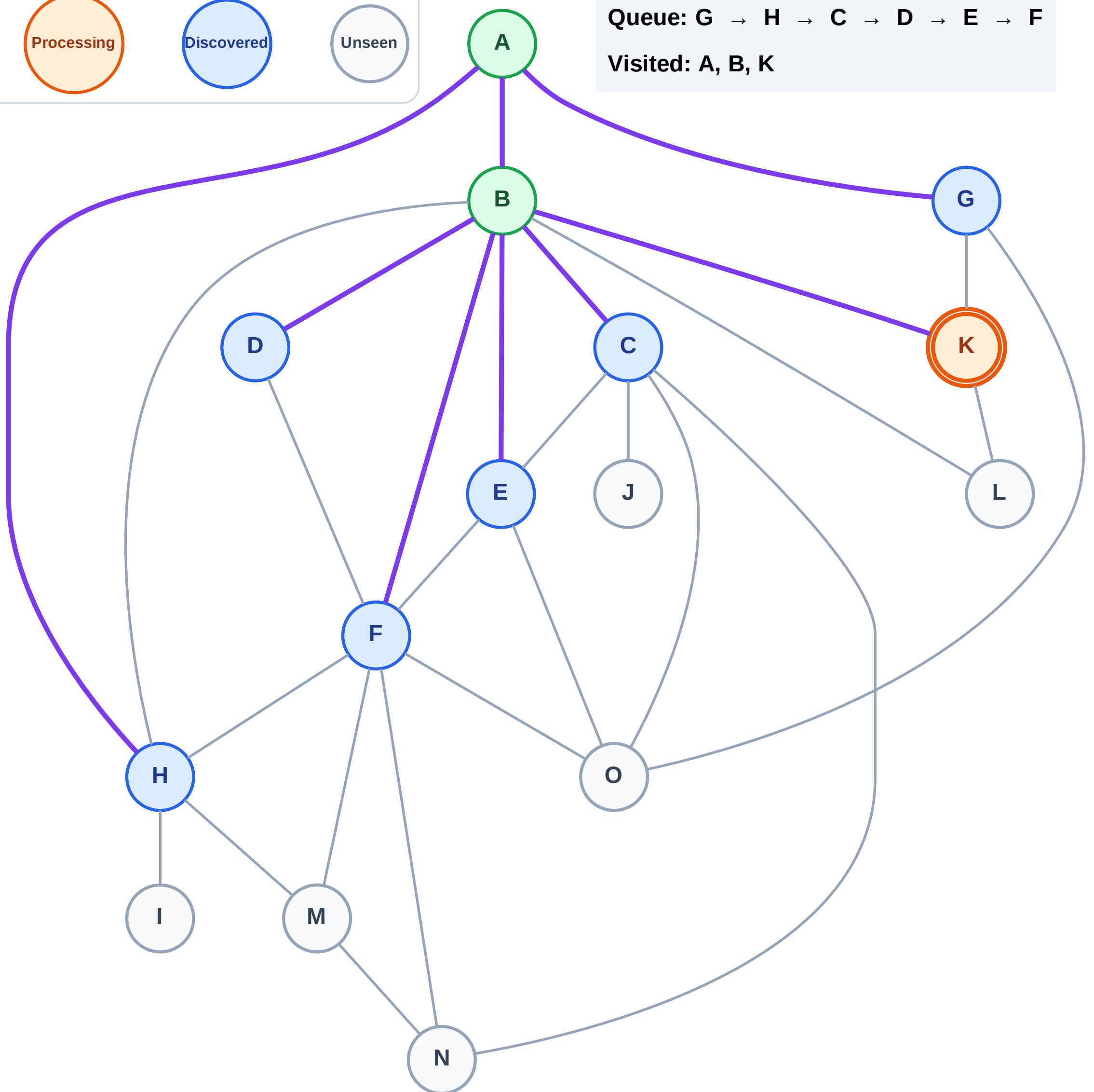
BFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



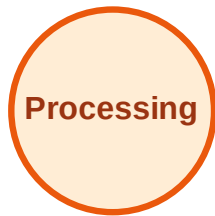
Queue: G → H → C → D → E → F
Visited: A, B, K



DFS Traversal

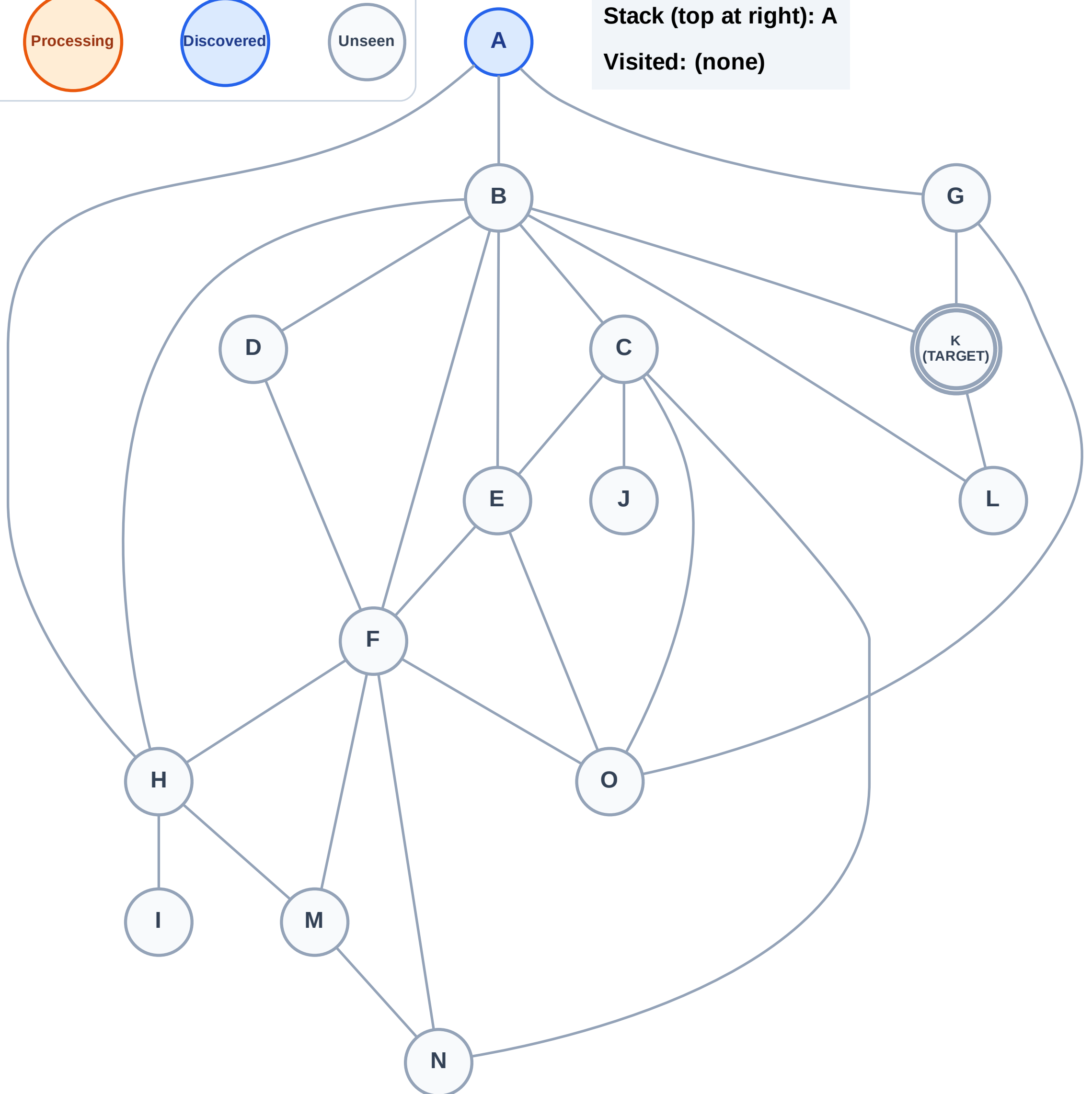
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

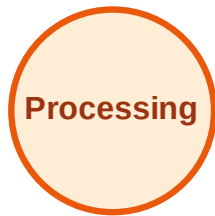
Visited: (none)



DFS Traversal

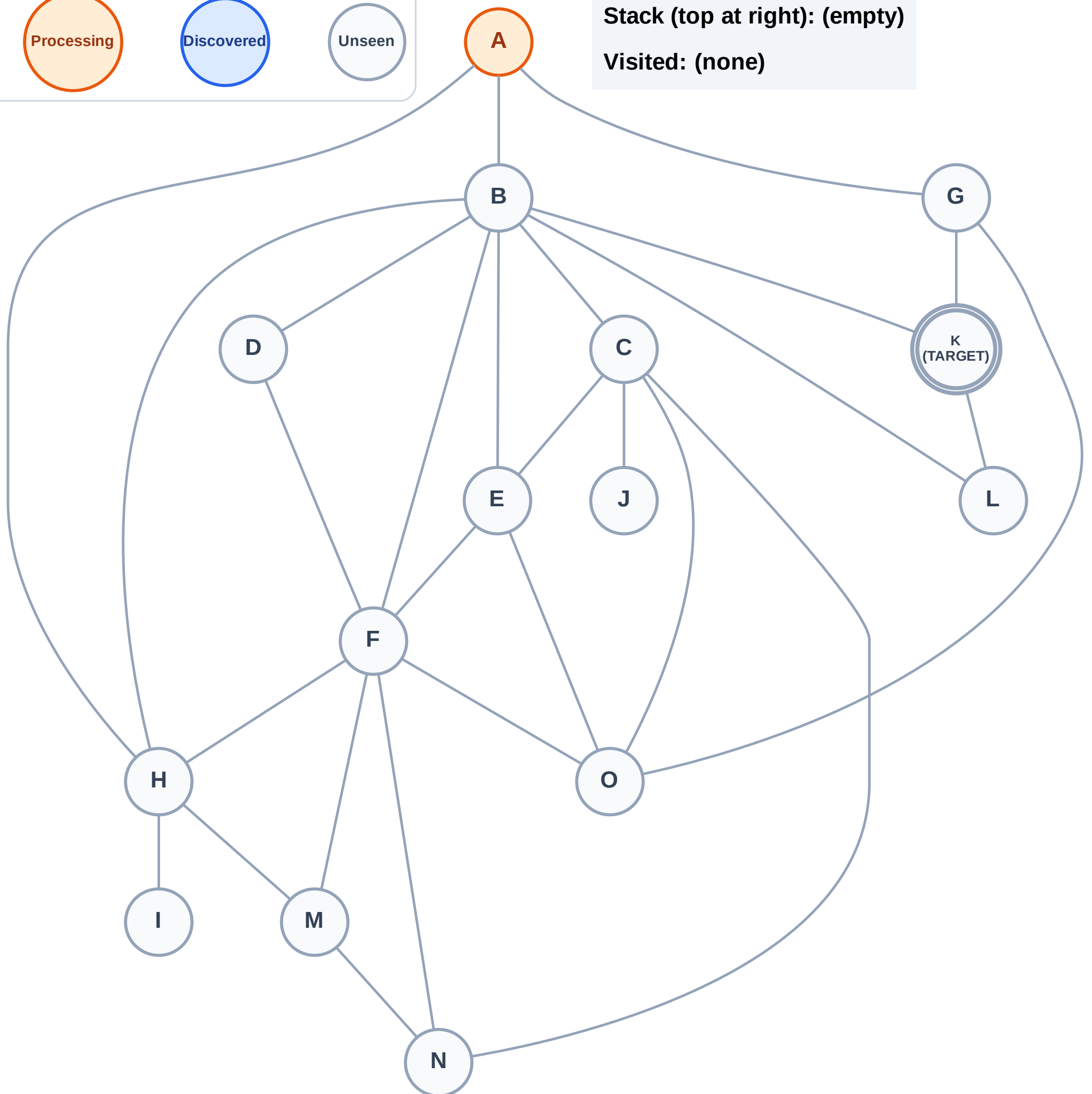
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



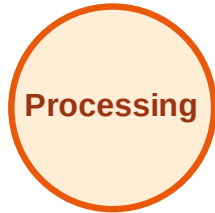
DFS Traversal

Step 3: Discover H, G, B from A

Legend



Complete



Processing



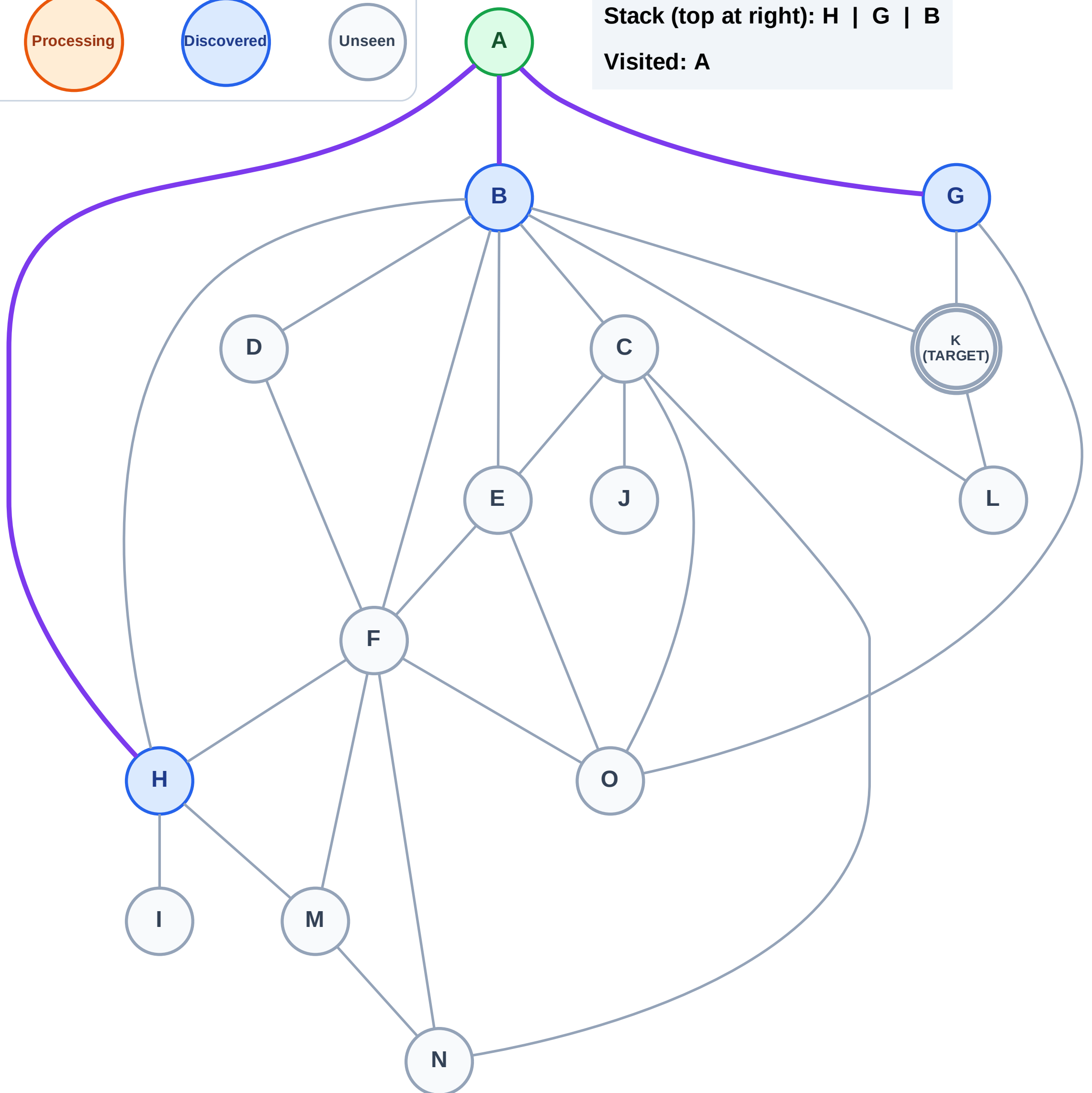
Discovered



Unseen

Stack (top at right): H | G | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

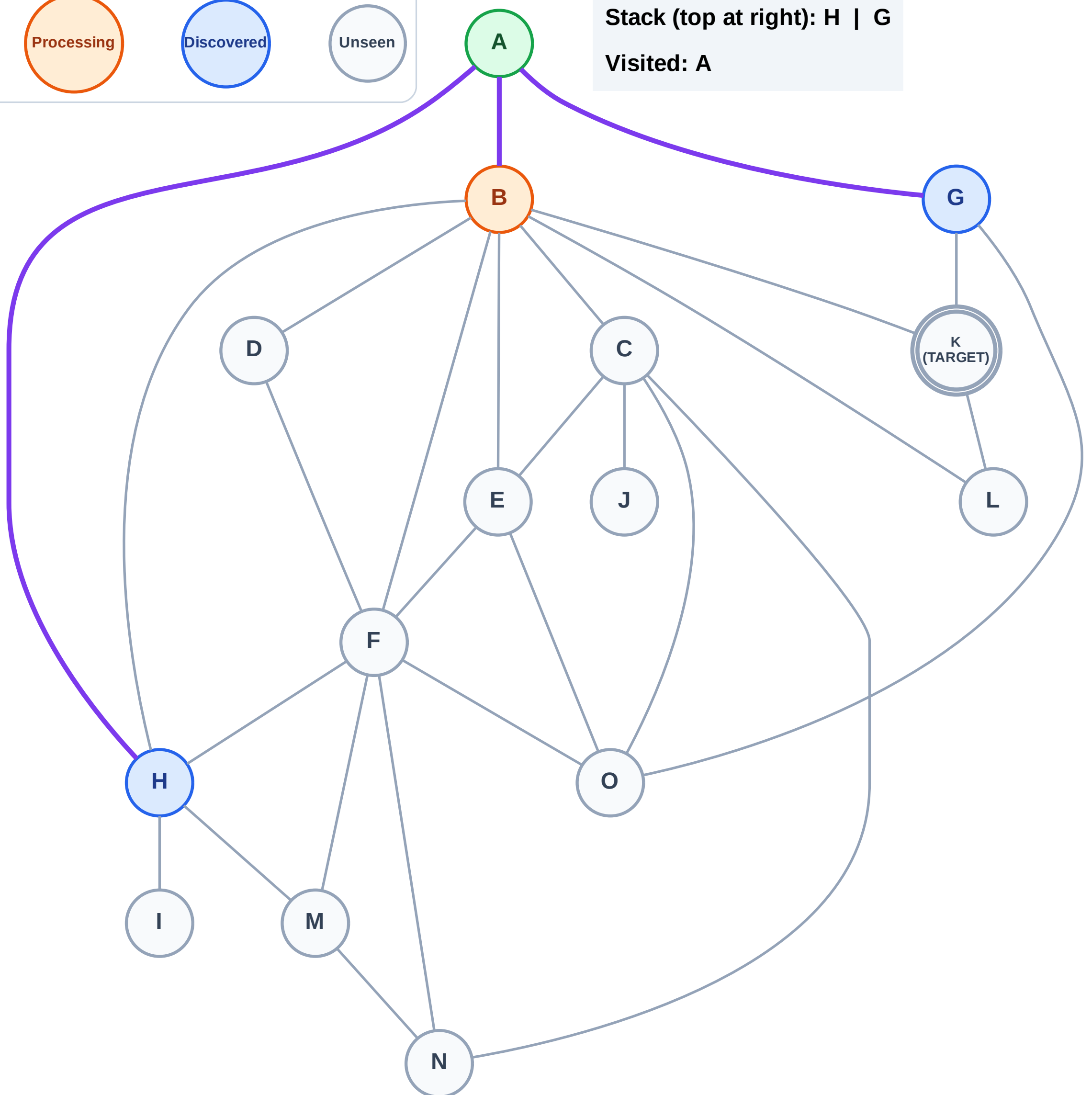
Processing

Discovered

Unseen

Stack (top at right): H | G

Visited: A



DFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



Stack (top at right): H | G | L

Visited: A, B, K

